

How Far Off the Recorded Path Can a Gaussian-Splat Driving Simulator Be Trusted?

CS 566 Project Proposal (one document per team) · Kyle Sha (ID 9085525476) · Allen Xia (ID 9085517234) · 29 Sep 2026

Main papers: [1] **Difix3D+** (CVPR 2025 Oral), *re-implemented* · [2] **FaithFusion** (CVPR 2026), *compared* · [3] **HUGSIM** (TPAMI 2026), *simulator for evaluation*

1. Problem Statement

Driving software is increasingly evaluated in simulators reconstructed from real driving logs: the recorded video becomes a **3D Gaussian splat** [13], a scene of millions of small, semi-transparent coloured ellipsoids that can be rendered from any camera position, so the simulated vehicle can leave the recorded trajectory. However, **a reconstruction is reliable only near that trajectory**; after a lane change or with a higher camera (e.g., a truck), rendered images degrade, and a driving system tested on them may behave differently than in reality. This project addresses two questions: **how far from the recorded trajectory a splat-based simulator remains trustworthy, and whether current generative repair methods ("fixers") restore trustworthiness or merely improve visual realism**. We define **trust** as the agreement between the outputs of a lane detector and a vehicle detector on a rendered image and on the **real** image captured at the same pose.

2. Significance and Motivation

Simulation is a principal means of validating driving software at scale; if off-trajectory renders are unreliable, a simulator may accept unsafe behaviour or reject safe behaviour. Both authors are members of Wisconsin Autonomous, a student autonomous-vehicle team, where we develop and test driving software in conventional simulators (MATLAB/Simulink, RoadRunner, CARLA). Their hand-built 3D assets differ visibly from real camera imagery, which limits how far results transfer to the vehicle. Reconstruction-based simulators promise to close this gap, but only if renders stay faithful off the recorded trajectory, and a generative fixer can make an image more convincing while making it less accurate. Characterizing this trade-off motivates the project, which draws on camera geometry, 3D reconstruction, image restoration, and CNN-based perception.

3. State of the Art

- **Splat-based simulators:** HUGSIM [3] executes driving policies in reconstructed scenes; the EUVS benchmark [7] shows that current methods degrade under large viewpoint changes.
- **Diffusion-based fixers:** Difix3D+ [1] repairs off-trajectory renders with a single-step diffusion model and adds the repaired views back as training images for the 3D scene, reporting roughly half the FID (a realism score; lower is better) of prior methods. NVIDIA Fixer [4] applies the same approach to driving data, and FaithFusion [2] restricts which pixels a fixer may overwrite, using a per-pixel map of how well the reconstruction already supports each pixel.
- **Evaluation practice:** these methods are evaluated primarily with image-realism metrics (FID, LPIPS) or with shifted cameras for which no real image exists (PRG-Fusion [6]). Studies with offset analysis or real off-trajectory images [9, 10, 11] neither compare fixers nor quantify hallucinated content; **to our knowledge, none compares fixers against real off-trajectory images in terms of the content they invent**.

4. Proposed Approach

- **Re-implementation:** we re-implement Difix3D+ [1], comprising the fixer's training procedure and its progressive update, which adds repaired views back as training images for the 3D scene. As in the paper, the fixer is fine-tuned from the pretrained SD-Turbo diffusion model; we write the training code ourselves from the paper, validate it against NVIDIA's official code and weights, and train it on driving data. A preliminary version already runs (§5).
- **New contributions:** (a) an **evaluation against real ground truth**, using Para-Lane [8], which scans the same roads in three parallel lanes (approximately 3 m and 6–7.7 m apart), and MV2 [9], in which a car and a scooter record the same road at different camera heights; training pairs are built from neighbouring lanes, each pairing a render moved into lane B with the real lane-B image (cf. ReCamDriving [12]), and are used to train our Difix and our lightweight fixer; (b) **hallucination accounting**, measuring invented or erased lane markings and spurious vehicles in addition to image similarity; (c) a **lightweight fixer** trained from scratch on the same pairs (15M parameters, about 60× smaller than Difix's diffusion network); (d) a **geometry-only road-surface prior**, re-implemented from the ViSE paper [15] (no code released), which forces the 3D road to lie on a smooth surface fitted to LiDAR and cannot introduce new content; and (e) a **cross-dataset transfer study** on US city scenes (EUVS benchmark [7], built on the MARS dataset) and Indian roads (MV2 [9]).

5. Limitations of Existing Approaches and Expected Advantages

Existing fixers are diffusion models trained to produce realistic images; they are therefore rewarded for plausible detail even where that detail is incorrect. A preliminary study (September 27–28) is consistent with this: at approximately 3 m lateral offset, **lane agreement** (the F1 score between lanes detected by CLNet on the render and on the real image) falls from 0.98 to 0.07, and

NVIDIA's released Difix improves image similarity (LPIPS $0.52 \rightarrow 0.44$) but not **lane-marking precision** (the fraction of painted-marking pixels in the render that match real markings; $0.52 \rightarrow 0.48$). We test three hypotheses. **(H1)** Training a fixer on real adjacent-lane pairs makes it more faithful, not only more realistic; in the preliminary study, our re-implemented Difix raised lane agreement from 0.07 to 0.71 on two held-out roads, at the cost of more **spurious vehicles** (vehicles detected in the render with no counterpart in the real image: 0.18 per image, versus 0.04 for the unrepaired render). **(H2)** A geometry-only prior improves lane fidelity without introducing hallucinated content. **(H3)** Fixers trained on one city do not transfer to others, so trustworthiness must be established per domain.

6. Evaluation Plan and Expected Results

- **Deviation axes:** lateral offset (real images from adjacent lanes at ≈ 3 m and 6–7.7 m); camera height (real car and scooter views from MV2; fallback: the real image re-projected onto the road plane at the new height); and heading (secondary: a pure rotation lets the real image be warped exactly to the new view, so only pixels the original camera saw are scored).
- **Metrics:** image similarity to the real image (PSNR and LPIPS, standard pixel and perceptual scores); lane-marking precision and recall (invented versus erased markings); lane-detector (CLRNet [14]) and vehicle-detector (YOLO11) agreement, following the metrics of ReconDreamer [5]; spurious-vehicle count; and, as an extension, closed-loop evaluation, in which a released driving policy (LTF) drives inside the HUGSIM simulator [3] and is scored on safety and progress.
- **Expected results** (all methods of §4 and the raw render): *Fig. 1*, trust vs. lateral offset; *Fig. 2*, trust vs. camera height; *Table 1*, hallucination statistics; *Table 2*, cross-dataset transfer; *Table 3*, our Difix vs. the official code; side-by-side videos with hallucinations annotated.

Components and how each is used:

Component (what it does)	Source	Our use
Difix image fixer [1]: a diffusion model that takes a flawed rendered image and outputs a cleaned-up version	NVIDIA (code, weights)	Re-implemented (our training code) and trained on our pairs
Progressive 3D update [1]: cleaned images are added back as training views, so the 3D scene itself improves off the recorded path	Difix3D+ paper	Re-implemented
FaithFusion [2]: limits which pixels a fixer may change, using a per-pixel confidence map	Released code	Comparison method (run as released)
NVIDIA Fixer [4]: NVIDIA's Difix trained on driving data	Released weights	Comparison method (run as released)
Lightweight fixer: small image-to-image U-Net (15M parameters)	—	Ours, trained from scratch
Road-surface prior [15]: fits a smooth road surface to LiDAR points and flattens the road in the 3D scene; involves no image generation	ViSE paper (no code)	Re-implemented
3D reconstruction [13]: builds each splat scene from video	gsplat library	Run by us per scene
Detectors and simulator: CLRNet lanes [14], YOLO11 vehicles, HUGSIM [3] with the LTF policy	Released, pretrained	Used as-is (evaluation only)

7. Timeline

Dates	Kyle Sha	Allen Xia
9/29–10/12	MV2 reconstruction and camera-height evaluation; additional Para-Lane test scenes	Evaluation framework for MV2 and EUVS; per-scene variance analysis
10/13–10/29	Validation of the Difix re-implementation against the official code	Road-surface prior on all scenes; hallucination analysis; mid-term report 10/29
10/30–11/20	Cross-dataset transfer study; progressive update on all scenes	Closed-loop evaluation in HUGSIM (extension)
11/21–12/10	Final figures and videos	Presentation (12/1–12/10) and project webpage (12/10)

Risks: preliminary results rest on few test roads (more scenes and per-scene variance will be reported); closed-loop tests use only LTF, as other policies need outdated dependencies.

References

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12. ReCamDriving, ECCV 2026 · arXiv 2512.03621
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